MID-SEMESTER PROGRESS REPORT

Time-Series Data Analysis and Trend Discovery in Pakistan's Crop Prices

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Course	Data Mining	**Deliverable**	Deliverable 1

1. Project Overview

Agricultural price volatility is a persistent challenge for farmers, policymakers, and market participants in Pakistan. Prices of essential crops fluctuate due to a complex interplay of seasonal cycles, supply-demand dynamics, and climatic factors. This project addresses this challenge through time-series data mining applied to the Crop Prices Dataset of Pakistan (Kaggle), containing historical price records across multiple markets with attributes including crop type, city, date, and price.

The core objective is to systematically discover and model seasonal patterns, long-term trends, and abnormal price fluctuations using a structured pipeline: data engineering, exploratory analysis, time-series decomposition, and forecasting using ARIMA and Holt-Winters Exponential Smoothing. Model accuracy is evaluated with MAE, RMSE, and MAPE on a chronologically correct train-test split, aiming to produce interpretable insights that support agricultural planning and market forecasting.

2. Dataset at a Glance

Total Records	Unique Cities	Unique Crops	Source Files
~7.99 Million	**138**	**76**	**53 CSV Files**
Date Range	**Avg Price (PKR)**	**Price Std Dev**	**Max Price (PKR)**
2008 – 2024	**5,475**	**6,484**	**97,850**

3. Milestones Achieved

M1 — Data Loading & Merging

Loaded and validated all 53 CSV files, applied schema checks, date parsing, numeric coercion, and merged them into a single unified dataset of ~7.99 million records, sorted chronologically per crop-city pair.

M2 — Preprocessing & EDA

Applied IQR-based outlier filtering, duplicate removal, calendar feature extraction, and StandardScaler normalization. Nine visualizations were generated covering distributions, seasonality, volatility, and cross-city/crop comparisons.

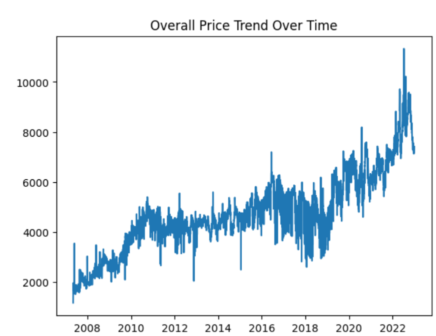

Fig 1. Price Trend Over Time

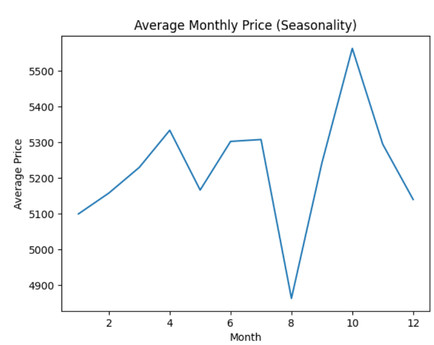

Fig 2. Monthly Seasonality Pattern

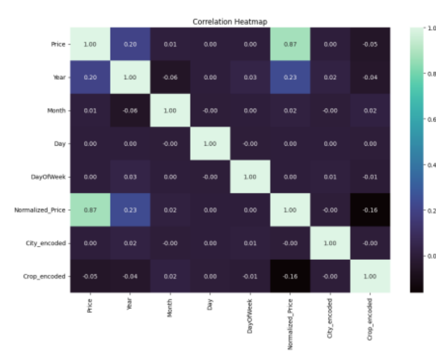

Fig 3. Feature Correlation Heatmap

M3 — Basic Time-Series Analysis

Selected Banana (DOZENS) in Vehari as the most data-rich series. Applied 7-day and 30-day moving averages alongside rolling statistics (mean and std) to characterize short-term trends, price stability, and volatility.

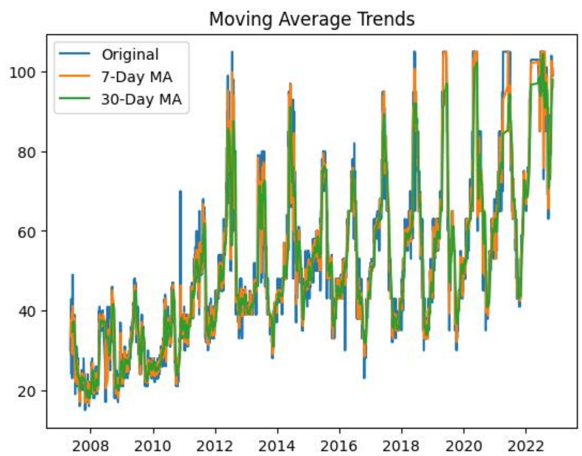

Fig 4. Moving Averages (7-Day & 30-Day)

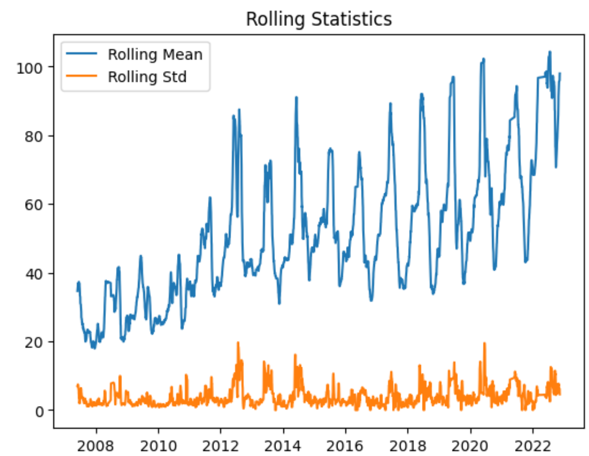

Fig 5. Rolling Mean & Std Dev (Volatility)

M4 — Advanced Time-Series Analysis

Extended analysis to the top 5 crops. Applied additive seasonal decomposition, ADF stationarity testing, and first-order differencing where required. Banana (DOZENS) and Garlic (China) were non-stationary and required differencing; the remaining three were stationary.

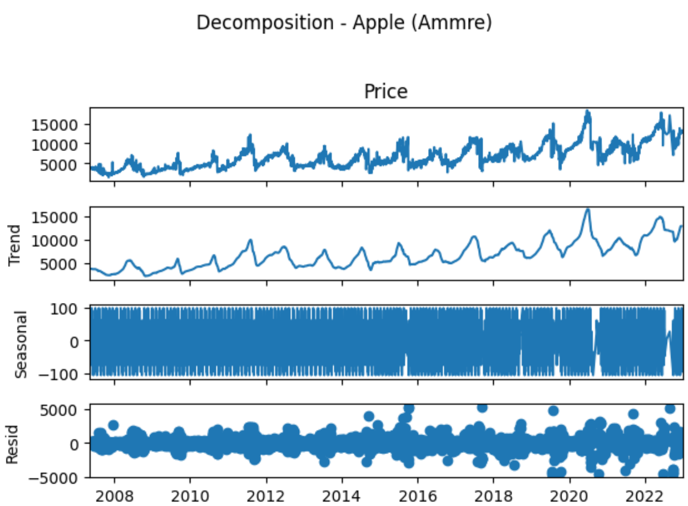

Fig 6. Seasonal Decomposition (Trend / Seasonal / Residual)

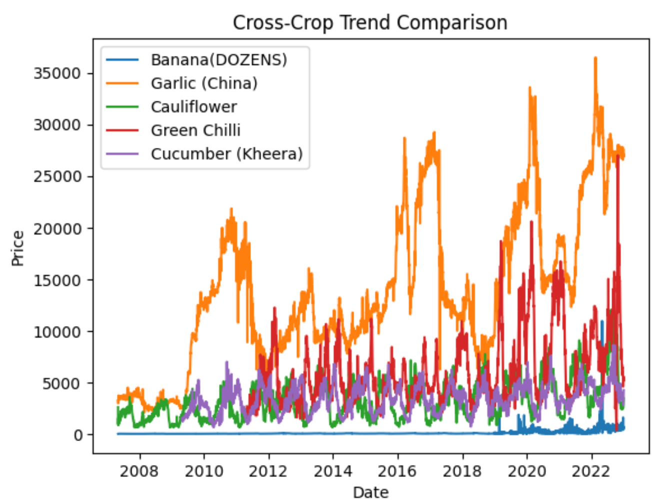

Fig 7. Price Trends Across Top 5 Crops

M5 — Temporal Feature Engineering

Built a model-ready feature matrix with 18 features: lag values (1, 3, 7, 14 days), rolling statistics, EMA, momentum features, cyclical month encoding, volatility, and group identifiers. Scaled with StandardScaler.

4. Key Findings So Far

- Prices exhibit strong monthly seasonality — several crops spike consistently in summer months, likely driven by supply-side climatic constraints.
- Green Chilli and Cucumber show the highest price volatility across cities, with rolling standard deviations significantly above the dataset mean.
- ADF testing revealed that 3 out of 5 top crops (Cauliflower, Green Chilli, Cucumber) are stationary in their original form, while Banana and Garlic required first-order differencing to achieve stationarity.
- The cross-crop trend comparison shows Garlic (China) has the steepest long-term upward price trend, while Cauliflower prices remain comparatively stable over the observed period.

5. Challenges Faced

- Irregular recording intervals — not all crop-city combinations have daily entries, creating gaps that complicate rolling window calculations and require careful forward-fill strategies.
- Price entries of zero were present in the raw data, requiring explicit filtering as they skew normalization and statistical summaries rather than representing genuine market values.
- Non-stationarity in several series — stationarity could not be assumed globally; ADF testing had to be applied per crop, and differencing applied selectively, adding complexity to the pipeline.

6. Progress Checklist

Task	Deliverable	Status
Dataset loading, validation & merging (53 CSV files)	Deliverable 1	✓ Done
Missing value handling & outlier removal (IQR)	Deliverable 1	✓ Done
Calendar feature extraction & price normalization	Deliverable 1	✓ Done
Exploratory data analysis (9 visualizations)	Deliverable 1	✓ Done
Basic time-series: moving averages & rolling statistics	Deliverable 1	✓ Done
Adv. time-series: decomposition, ADF test, differencing	Deliverable 1	✓ Done
Cross-crop trend comparison (top 5 crops)	Deliverable 1	✓ Done
Temporal feature engineering (18 features, scaled matrix)	Deliverable 1	✓ Done
ACF/PACF analysis & ARIMA model training	Final	☐ Pending
Holt-Winters Exponential Smoothing implementation	Final	☐ Pending
ML-Models: Linear Regression & Random Forest	Final	☐ Pending
Evaluation: MAE, RMSE, MAPE on temporal train-test split	Final	☐ Pending
Hyperparameter tuning & residual analysis	Final	☐ Pending
Clustering Analysis (if required)	Final	☐ Pending
Final comparative evaluation & trend insight extraction	Final	☐ Pending
Complete analytical report & visualizations	Final	☐ Pending